CURBCOUNT · STEM CHECK

CurbCount STEM check

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/01b-stem-check.md @ 82b4850		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Kit 1.9.1 upgrade: first STEM check of the TRL 3 design and of option B; one fix under the standing instruction (S22, S23), four fixes proposed to Amish (S5, S8, S26, S28); independent review applied

CurbCount STEM check

Verdict: the working principle (a coarse thermal array that cannot resolve a face, counting warm blobs by class and direction) sits within known science, and so does option B (60 GHz radar), but four details fail and are open. Of the 31 claims checked, **13 are within known science, 14 are plausible and 4 fail with a fix proposed; none fails outright.** The four open failures all touch the shared FieldNode core, so they go to Amish: S5 (a car's warm hood is in the field for 3.2 frames, not the 4 needed; fix: 32 Hz subpages on a 1 MHz bus), S8 (LiFePO4 may not be charged below 0 degrees C, so the node is energy neutral only where the winter air stays above about 0 degrees C), S26 and S28 (option B: a shared cell above its charge limit on hot days, and a charger set for 1.5 A under a 20 W panel). Option B's first drawing also failed (S22 and S23: the radar running continuously drew 32 Wh/day against 14.5 Wh/day stored); the fix, decided under Amish's standing instruction of 2026-10-23, is duty-cycled frame bursts (CBC-DEC-001 C1), tightened to 0.2 W average after the independent review. `project.yaml` records `stem_check: open`. Method and rules: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`, which imports the sizing of CBC-CAL-001.

Claims checked

Table 1. Claims, governing science and verdicts (generated from `docs/04-calcs/stem_results.csv`).

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	The thermal array cannot resolve a face or a number plate from 4.3 m: under the IEC 62676-4 detection level at head height	Physics	Optics and radiation	Pixels per metre = 1 / (range x pixel angle); IEC 62676-4 levels detect 25, recognise 125, identify 250 px/m	7.64 px/m <= 25 px/m	Within known science	CBC-CAL-001 B2 (7.6 px/m, 8.5 for 2.0 m)
S2	One pole covers a 3 m sidewalk, a 2 m bike lane and the nearest 3.5 m lane (110 degree axis across the street)	Physics	Optics and radiation	Ray geometry of the 110 x 75 degree lens, tilt 7.5 degrees toward the road	10.9 deg margin on the sidewalk side >= 0 deg margin on the sidewalk side	Within known science	CBC-CAL-001 A2, A3 (10.9 deg margin on the sidewalk side)
S3	A person stands out from pavement by more than the single-frame detection contrast on a mild design day	Physics	Heat transfer	Surface energy balance and the array's noise: NETD 0.1 K RMS at 1 Hz (Melexis MLX90640 datasheet) scales as the square root of the refresh rate; 4 sigma on a person that fills 0.37 of a pixel behind a 0.75 transmission window	smallest contrast 5.3 K; blind 10.2 h a day on sunlit pavement with single frames, 3.2 h with track averaging, 0.0 h shaded (assumed surface temperatures, to measure)	Plausible, to confirm at TRL 3	Open
S4	A 0.5 mm unpigmented HDPE film passes at least 0.5 of 8 to 14 um radiation	Physics	Optics and radiation	Two Fresnel reflections (n 1.53) and bulk absorption of 4 per cm, band-averaged (assumed; polyethylene is known to be IR transparent, thin film transmits most of the band)	0.75 transmission >= 0.5 transmission	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S5	A car at 50 km/h stays in the field for the 4 frames the tracker needs	Engineering and technology	Sensing resolution	Frames = field length / speed x frame rate (8 full frames a second from 16 Hz subpages). CAL-001 C1 treats the car as a point at road level (4.2 frames); the warm hood is about 1.0 m up, where the along-street field shrinks by (4.3 - 1.0) / 4.3 (independent review)	4.2 frames at road level, 3.2 at the hood, against 4: fails. Fix proposed: 32 Hz subpages (16 frames a second) give 6.4, at 1 MHz I2C (bus 24 % busy at 16 Hz, 48 % at 32 Hz, CAL-001 F1), noise up by the square root of 2 (recovered by track averaging) and a faster pull-up (644 ohm, F2); this changes the sensor port speed of the shared FieldNode core, so it goes to Amish (CBC-DEC-001 O3)	Fails, fix proposed	Open
S6	Winter solar harvest covers the daily load (6 W panel, 1.5 peak sun hours, 40 % derating)	Physics	Electromagnetism and circuits	Stored energy = panel W x sun hours x derating x MPPT efficiency x charge efficiency	4.36 Wh/day >= 2.47 Wh/day	Within known science	CBC-CAL-001 E3 (4.4 Wh against 2.47 Wh, while the cell can charge: see S8, S25)
S7	One cell carries the node for 3 days without sun	Chemistry	Electrochemistry and batteries	Usable energy = 3.2 V x 6 Ah x 80 % depth of discharge; load 103 mW average; 70 % capacity at -20 C and 80 % at end of life (independent review: both together)	3 days (-20 C, end of life) <= 3.49 days (-20 C, end of life)	Within known science	CBC-CAL-001 E2 (4.36 days at -20 C; 3.48 days with the end-of-life 80 % as well)
S8	The node stays energy neutral	Chemistry	Electrochemistry and batteries	LiFePO4 cells must not be charged below 0	by this estimate the cell can charge only when	Fails, fix proposed	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	through a winter at sites where the air falls below freezing			C (lithium plating); FieldNode's NTC lockout stops charging. First-order enclosure temperature in winter sun: (0.4 x 300 W/m2 x 0.05 m2 x 0.6 shade + 0.1 W) / 1.2 W/K = 3.1 K above the air (assumptions)	the air is above about -3 C, and nearer 0 C allowing for the thermal lag of the cell after a cold night (independent review), so 0 C is taken as the limit; below that the node lives on the stored 4.4 days and then stops, so energy neutrality holds for cool sites, not for a Toronto January. The claim fails for cold sites; fixes (heated cell pocket, low-temperature cell) change the shared FieldNode core and the scope, so they go to Amish (CBC-DEC-001 O1)		
S9	The 10-byte record fits US915 DR0 and its airtime stays under the 400 ms dwell limit	Engineering and technology	Power and link budgets	LoRa time on air (Semtech AN1200.13) for 10 + 13 bytes at SF10, 125 kHz; US915 DR0 allows 11 bytes	371 ms <= 400 ms	Within known science	CBC-CAL-001 G2 (371 ms)
S10	Uplinks stay inside the EU868 1 % duty cycle	Engineering and technology	Power and link budgets	Airtime per hour = 4 x time on air at SF9 (206 ms)	0.82 s/h <= 36 s/h	Within known science	CBC-CAL-001 G1 (0.82 s/h)
S11	Six 12-bit counters hold a 15-minute count in the busiest lane	Mathematics	Orders of magnitude	Nearest lane at 1,800 vehicles/h (saturation flow, assumed) is 450 per 15 min; a 12-bit counter holds 4,095	450 counts per bin <= 4095 counts per bin	Within known science	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S12	Wind load: the post carries the panel at 35 m/s gusts with a safety factor of 1.5	Physics	Strength of materials	Drag = 0.5 rho v2 Cf A; bending of a 42.4 x 3 mm aluminium post, 6063-T6 yield 150 MPa	2.7 MPa <= 100 MPa	Within known science	CBC-CAL-001 H1 (stress 2.7 MPa class, factor 55)
S13	The band saddles hold the arm against the twist of the wind on arm and head	Physics	Mechanics and statics	Friction torque of two bands at 1,000 N preload each, mu 0.2 (assumed) against the wind torque; if the liners creep and half the preload is lost the factor falls to about 1.6 (independent review), so a re-torque interval is needed	14 N m <= 30.5 N m	Within known science	CBC-CAL-001 H3 (factor 3.3)
S14	The counter weighs 6 kg or less on the pole	Engineering and technology	Manufacturing tolerances	Sum of modelled volumes times density plus catalogue masses of bought parts	5.24 kg <= 6 kg	Within known science	CBC-CAL-001 H5
S15	The I2C bus runs at 400 kHz over the 2 m M12 cable	Physics	Electromagnetism and circuits	Rise time 0.8473 R C <= 300 ns (I2C fast mode) with 220 pF of cable (100 pF/m assumed, to measure): pull-up of at most 1.6 kohm, sink current 2 mA (within the 3 mA limit)	1.5 kohm pull-up <= 1.61 kohm pull-up	Plausible, to confirm at TRL 3	Open
S16	The STM32WL runs the fixed-point tracker (about 150 cycles per pixel, 2 % of a 48 MHz core,	Engineering and technology	Control	Operation counts: 768 pixels x 150 cycles x 8 frames/s / 48 MHz; RAM tally of CBC-CAL-001 F4 (64 kB part); the	cycle count is an estimate until the tracker is profiled on the part (TRL 4, on hold)	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	about 40 kB of RAM)			vendor's floating-point conversion would not fit (50 % of the core)			
S17	Parts cost stays within the USD 275 value-engineering target	Engineering and technology	Standards practice	Sum of the 14 BOM lines (indicative prices; thermal array USD 64.95 at SparkFun, checked 2026-10-31)	274 USD <= 275 USD	Plausible, to confirm at TRL 3	Open
S18	The window film survives five years of sun	Chemistry	Materials and polymer chemistry	Unstabilised polyethylene photo-oxidises and embrittles under UV; carbon black would stop it but also blocks the infrared, so a UV-stabilised (hindered amine) clear grade is needed	no grade chosen; film to be specified and measured (TRL 4, on hold)	Plausible, to confirm at TRL 3	Open
S19	Aluminium saddles, plates and arm with stainless bands and bolts last five years on a steel street pole	Chemistry	Inorganic	Aluminium is anodic to stainless steel and to a steel pole in a wet, salted street; the contacts are small and painted or coated in practice, but the joints are not specified	isolating washers or coating to specify; to confirm in the build plan checks (TRL 4, on hold)	Plausible, to confirm at TRL 3	Open
S20	The counter records no image and no personal identifier: only six counters leave the device in normal operation	Biology	Human physiology and biomechanics	No personal data can be recovered from 7.6 px/m thermal frames held in RAM and overwritten every 125 ms; the record identifies no one (data protection by design, GDPR	S1 gives the resolution; the firmware path for the jumper stream, its limits and the small-count bins are firmware design and audit questions (TRL 4, on hold)	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				Article 25). During an aim check the calibration jumper streams frames to a laptop at the pole (CBC-CAL-001 H6, PR-020), and quiet night-time bins can single out one person (independent review)			
S21	Option B: RF exposure of a person under the radar is far below the ICNIRP limit	Biology	Toxicology	Power density = EIRP / (4 pi r2) at head height, 2.40 m, with 1 W EIRP (a deliberate overestimate); ICNIRP 2020 general public limit 10 W/m2 above 6 GHz (to confirm against the guideline text)	13.82 mW/m2 <= 1e+04 mW/m2	Within known science	Open
S22	Option B: a 20 W panel keeps the radar head energy neutral in winter (radar frames duty-cycled to 0.2 W average; running continuously at 1.2 W it drew 32 Wh/ day and failed)	Physics	Electromagnetism and circuits	Stored energy of a 20 W panel (1.5 peak sun hours, 40 % derating) against the daily draw; fix decided under the standing instruction (CBC-DEC-001 C1): one frame burst in six. TI's reference may draw more than 1.2 W and the burst rate the tracker needs is unproven (independent review)	5.3 Wh/day <= 14.5 Wh/day stored on paper; the burst rate that still gives 4 frames per vehicle is to be profiled (TRL 4, on hold)	Plausible, to confirm at TRL 3	Open
S23		Chemistry					

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	Option B: two cells carry the radar head for 3 days without sun (0.2 W duty-cycled average; at 0.25 W they gave only 2.6 days with cold and end of life together and failed)		Electrochemistry and batteries	Usable energy of two 3.2 V 6 Ah cells at 80 % against the duty-cycled draw, 70 % capacity at -20 C and 80 % at end of life, together (independent review)	3 days (-20 C, end of life) <= 3.22 days (-20 C, end of life)	Within known science	CBC-DEC-001 C1
S24	Option B: a 60 GHz radar counts people and vehicles by range, angle and speed in rain, dark and heat	Physics	Optics and radiation	FMCW radar resolves range from the beat frequency and speed from phase; Texas Instruments offers a people counting and tracking reference for the IWR6843 (TI, checked 2026-10-31)	the principle is established; counting accuracy and false counts from wind-blown foliage are not shown (TRL 4, on hold)	Plausible, to confirm at TRL 3	Open
S25	Option A at temperate sites: the cell stays below its 45 C charge limit	Chemistry	Electrochemistry and batteries	Enclosure temperature = air + 3.1 K (S8 estimate) with the temperate design day capped at 20 C air (PR-008); LiFePO4 charging allowed 0 to 45 C (cell datasheet class)	23 C <= 45 C; on a hot clear day outside this scope FieldNode's own figures give 0.8 Wh stored against 2.47 Wh drawn without a shield (CBC-CAL-001 E6)	Within known science	CBC-CAL-001 E6 (temperate scope, PR-008)
S26	Option B at hot sites: the cell can still charge when the air is near 40 C	Chemistry	Electrochemistry and batteries	LiFePO4 charging is allowed to 45 C only (cell datasheet class); FieldNode's own figures put the cell above the limit for most sun hours on a hot	option B is the hot-climate variant but shares the same cell and enclosure, so it needs the shield or a cell pack in shade; a shared-core change, so it	Fails, fix proposed	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				clear day and store 0.8 Wh against 2.47 Wh drawn without a shield, 13.1 Wh with its proposed shield (CBC-CAL-001 E6)	goes to Amish with option B (CBC-DEC-001 O4)		
S27	Option B stays at 6 kg or less on the pole	Engineering and technology	Manufacturing tolerances	Option A's 5.24 kg plus the CAD mass difference between the two models (0.4 kg: second cell, larger panel, radar head); the independent review put it nearer 6.8 kg, to profile	5.64 kg <= 6 kg	Plausible, to confirm at TRL 3	Open
S28	Option B: FieldNode's charger passes the 20 W panel's current into two cells	Physics	Electromagnetism and circuits	Peak charge current of a 20 W panel at 1,000 W/m2 is 5.0 A (CBC-CAL-001 E5) against FieldNode's standard charger setting of 1.5 A; two cells in parallel take 2.5 A each (0.42 C)	the charger setting caps the harvest unless FieldNode's high-load variant (raised in CBC-DDR-001 D3) is used; a shared-core change, so it goes to Amish with option B (CBC-DEC-001 O4)	Fails, fix proposed	Open
S29	The window stays clear of dew, frost, insects and droppings	Physics	Optics and radiation	A film of water or frost on the window absorbs 8 to 14 um radiation strongly (a 10 um water film is estimated to pass about 0.37 by the independent review; not recomputed here); webs, insects and	no heater, hydrophobic coating or wiper is in the design; contrast margin (S3) and cleaning interval to measure (TRL 4, on hold)	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				droppings block pixels			
S30	The 2 m sensor cable and the node survive surges and electrostatic discharge on a street pole	Engineering and technology	Standards practice	A cable running up a pole is exposed to induced surges and static discharge; the BOM has resettable fuses (line 14) but no surge protection devices	protection to specify with the FieldNode project; to assess (TRL 4, on hold)	Plausible, to confirm at TRL 3	Open
S31	The tracker classes people, bicycles and cars correctly and does not count dogs, umbrellas or prams as extra people	Engineering and technology	Control	Class by blob area, speed and track length at 32 x 24 pixels; a dog or a pram is a small warm blob, an umbrella can hide a person from above (independent review)	no calculation possible at TRL 3; replay of recorded frames at TRL 4 (on hold); tracked in CBC-RSK-001 F9	Plausible, to confirm at TRL 3	Open

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**. A plausible row is closed by a CAL result or stays open until the test that settles it, which is TRL 4 work and on hold.

Fixes made and fixes proposed

Item	What failed	Fix	Result
Option B energy (S22, S23)	The 1.2 W radar running continuously draws 32.0 Wh/day against 14.5 Wh/day stored by a 20 W panel. Bursts at 0.4 W average gave 2.0 days at -20 degrees C; 0.25 W gave 3.2 days but only 2.6 with the end-of-life 80 % as well (independent review)	Decided under the standing instruction (CBC-DEC-001 C1): the radar runs in frame bursts, one in six, for about 0.2 W average; two cells and a 20 W panel stay as drawn	5.3 Wh/day against 14.5 stored; 5.8 days without sun, 3.2 days at -20 degrees C and end of life. The burst rate that still gives the tracker its 4 frames per vehicle is to be profiled (TRL 4), so S22 stays plausible
Frames per car (S5)	At the warm hood, 1.0 m up, the along-street field gives 3.2 frames at 50 km/h, against 4 assumed needed (CAL-001 C1 used road level, 4.2)	**Not implemented**: 32 Hz subpages give 6.4 frames, at 1 MHz I2C (48 % busy) and a pull-up of 644 ohm or less; it sets the sensor port speed of the shared FieldNode core, so it goes to Amish (CBC-DEC-001 O3)	Open; PR-001 stays at risk
Cold charging (S8)	LiFePO4 must not be charged below 0 degrees C, and FieldNode's NTC lockout stops charging. In shaded winter sun the box sits about 3 K above the air, so charging stops at about -3 degrees C, nearer 0 degrees C allowing for thermal lag	**Not implemented**: a heated cell pocket or a low-temperature cell changes the shared FieldNode core and the scope, so it goes to Amish (CBC-DEC-001 O1)	Open. Until decided, the claim is limited to sites where the winter air stays above about 0 degrees C (PR-010 at risk)
Option B at hot sites and charger (S26, S28)	The shared cell is above its 45 degrees C charge limit for most sun hours on a hot clear day (FieldNode's figures, CAL-001 E6); the 20 W panel pushes 5.0 A against the 1.5 A charger setting	**Not implemented**: a shield and FieldNode's high-load charger change the shared core; if option B is carried forward they go to Amish (CBC-DEC-001 O4)	Open, only for option B

Fields not relevant

None. Every field is touched: physics (optics, heat transfer, circuits, wind loads), chemistry (cell charging, polymer ageing, galvanic corrosion), biology (privacy of imagery, RF exposure under option B), mathematics (counter size) and engineering (frame counts, link budgets, processor load, cost).

Concept options

- Option A, thermal array head: rows S1 to S20, S25, S29 to S31. Two fail with a fix proposed (S5, S8); eleven are plausible and each names what settles it; eleven are within known science.
- Option B, 60 GHz radar head: rows S21 to S24 and S26 to S28, and S8 shared with A. Within known science once duty-cycled (S21, S23); S22, S24 and S27 are plausible; S26 and S28 fail with a fix proposed (a hot-site shield and FieldNode's high-load charger). No option fails outright, so neither is dropped.

Independent review

A second reviewer (a separate headless Claude session on the current model, `claude -p --model claude-opus-5-5`, because this session has no subagent tool) was given only this document, the calculation note CBC-CAL-001, the product requirements, `sizing.py` and `stem_check.py`, and none of the author's reasoning. It was asked to refute the headline results and every "Within known science" row. It reproduced every headline number (7.64 px/m, 10.9 degrees, 4.19 frames, 2.47 Wh/day, 4.36 days, 370.7 ms, 2.74 MPa, 12.24 mW/m2) and found the following; what changed is in the last column.

Item	Finding	Change
S1, S2 (pixel size, coverage)	Upheld. The claim holds for passers-by only: an installer within 0.5 m is above the detection level; coverage ignores occlusion by crowds and parked cars	PR-004 status text states passers-by; occlusion noted; the 0.4 m shift of the head is not adopted (no dimension changed)
S5 (frames per car)	**Refuted.** CAL-001 treats the car as a point at road level; at the warm hood, 1.0 m up, the field is shorter: 3.2 frames, not 4.2	S5 now "Fails, fix proposed" (32 Hz subpages and 1 MHz I2C give 6.4); not implemented because it changes the shared sensor port speed: CBC-DEC-001 O3; PR-001 text updated
S3 (contrast)	Weakened. The 0.1 K NETD is the 55 degree part; the 110 degree part is noisier	Left plausible; the 4 sigma row already carries an edge-of-lens noise factor 1.25 in CAL-001 D2; to measure
D4 (blind hours)	Upheld for the stated days; summer days above 20 degrees C are outside the claim	PR-008 states summer is outside the claim
S6 (winter harvest)	Weakened. The row says within while PR-010 is at risk through S8; a refill after three sunless days takes about 3.9 days	S6 closed_by now depends on S8 and S25
S7 (autonomy)	Upheld. Cold and end of life together give 3.49 days	S7 now uses both together
Hot-weather charging	Missing. The cell is above its 45 C charge limit for most sun hours on a hot clear day (CAL-001 E6)	S25 (temperate scope, within) and S26 (option B at hot sites, fails with fix proposed, O4) added
S8 (cold charging)	Weakened. The failure stands but -5 C is optimistic: the 5 K rise ignores street shade and the cell's thermal lag	Shade derating added (3.1 K, about -3 C); 0 C taken as the limit; text and PR-010 updated
S9 (US915)	Upheld. 96 uplinks a day at DR0 is 35.6 s/day, above The Things Network's 30 s/day fair-use guidance	Noted; fewer uplinks at DR0 or another network is a deployment choice
S10, S11	Upheld	None
S12 (wind on the post)	Upheld. The weak points are more likely fasteners and the rail plate bolts than the post wall	Noted for the design FMEA (F1, F2)
S13 (band twist)	Upheld but fragile: half the preload gives a factor of about 1.6	Text states it; a re-torque interval is needed (BLD-001, to confirm)
S14 (mass)	Weakened. The reviewer found 5.26 kg in the CAL text and 5.24 in the script	The script prints 5.24 kg and the text of the PRS now says so; the CAD model reads 4.8 kg and the two are not itemised against each other
S15 (I2C)	Weakened to plausible: 1.5 kohm against 1.61 kohm is a 7 % margin on an assumed 100 pF/m	S15 now Plausible
S20 (privacy)	**Refuted as stated.** The calibration jumper streams frames to a laptop; the claim is a firmware claim	S20 restated to normal operation, now Plausible, closed by a firmware audit (TRL 4)

Item	Finding	Change
S21 (RF exposure)	Upheld; range is 2.40 m for a person on the sidewalk, 13.8 mW/m2	Range corrected
S22 (option B energy)	Weakened to plausible: rests on an unproven burst rate	S22 now Plausible
S23 (option B autonomy)	**Refuted.** Cold and end of life together give 2.6 days at 0.25 W	Fix tightened to 0.2 W (one burst in six): 3.2 days; S23 within
Option B overall	Missing: the 20 W panel pushes 5 A against the 1.5 A charger; mass near 6.8 kg; wind on the panel; cost unchecked	S27 (mass, plausible: 5.64 kg here), S28 (charger, fails with fix proposed, O4) added; wind and cost remain open for option B
S17 (cost)	Weakened. A margin of 0.4 % is smaller than the price uncertainty	PR-016 is now "at risk"
Install time	Weakened. The task list leaves out platform set-up, traffic control and teardown (75 to 90 min in all)	PR-013 text updated
Missing physics, biology, engineering	Dew, frost, webs and droppings on the window; umbrellas; snow on the panel; head height against vehicle clearance; dogs; surges and static on the cable	S29, S30 and S31 added; clearance added to CBC-REG-001 G11; snow and umbrellas noted, not given rows
Missing mathematics	No uncertainty model for counting accuracy beyond random merging	Noted; a count-error model belongs with the field comparison (TRL 4, on hold)

The reviewer would downgrade S5, S23, S20, S22 and S15 and leave S3, S4, S16, S17, S18, S19 and S24; all are done above. Numbers it computed that are not recomputed in `stem_check.py` (the 12 h blind day, 0.37 water-film transmission, 3.9 day refill, 35.6 s/day) are stated as the reviewer's.

Sources

- Melexis, MLX90640 32x24 IR array datasheet (NETD 0.1 K RMS at 1 Hz, fields of view), https://media.melexis.com/-/media/files/documents/datasheets/MLX90640-datasheet-melexis.pdf, accessed 2026-10-31.
- SparkFun, IR Array Sensor 110 degree FOV MLX90640 (price 64.95 USD), https://www.sparkfun.com/ir-array-sensor-110-degree-fov-mlx90640.html, accessed 2026-10-31.
- Texas Instruments, IWR6843 single-chip 60 GHz to 64 GHz mmWave sensor, https://www.ti.com/product/IWR6843, and IWR6843ISK-ODS evaluation module (no price listed), https://www.ti.com/tool/IWR6843ISK-ODS, accessed 2026-10-31.
- IEC 62676-4 pixel density levels (detect 25, recognise 125, identify 250 px/m), as used in CBC-PRB-001 and CBC-CAL-001 B.
- ICNIRP, Guidelines for limiting exposure to electromagnetic fields (100 kHz to 300 GHz), *Health Physics* 118(5), 2020 (general public limit above 6 GHz; value used in S21 to confirm against the text).
- Regulation (EU) 2016/679, Article 25, https://eur-lex.europa.eu/eli/reg/2016/679/oj (S20).

- LiFePO4 charge limits: cell maker datasheet class (charge 0 to 45 degrees C), as used in BOM line 3 and FieldNode BOM line 6 (S8).
- CBC-CAL-001 v0.4 and `docs/04-calcs/sizing.py` for every row that cites a CAL item.